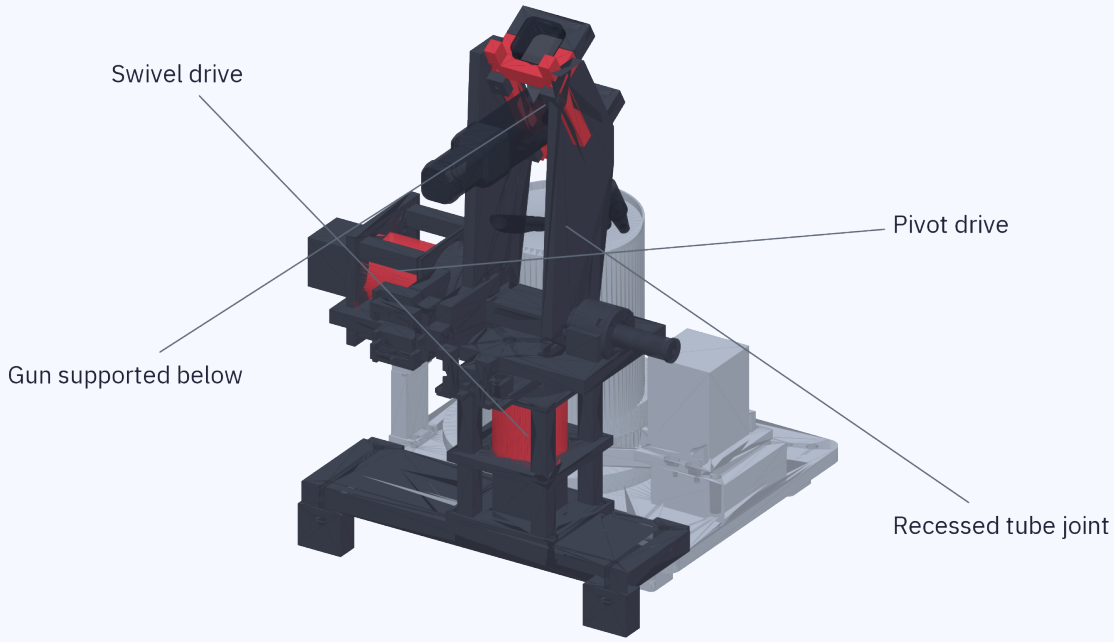
Gun positioner assembly guide

Two coaxial PGFUN 50:1 drives.
Swivel base, horizontal pivot, gun supported from below.



The fixture mounts to the existing rotator base. Coral identifies the two drives and lower housing cradle.

ONE COMPLETE BUILD

$479.36 mechanism/controller + $524.50 camera/lens = **$1,003.86** before tax. Prime prices observed October 5, 2026.

Printing, assembly, wiring and one-camera dry-run learning

Find the operation

Build in order, use the picture at each operation, and complete its visible check before continuing.

Build the positioner 3

Parts and fasteners 16

Wire the controller 19

Commission the fixture 22

Learn with one camera 25

Engineering reference 29

Firmware and host commands 31

Print manifest 33

READ THE PICTURES

Coral identifies the parts used by the operation. Dark grey is the printed fixture and gun; light grey is the existing rotator or purchased hardware. CAD dimensions and the print manifest govern the build.

SET THE WORKING POSE

The 60-degree attitude and 16 mm nozzle gap are editable print parameters. Establish welding focus and attitude on a sacrificial joint, then record the accepted setup. Use the supplied STL orientation.

Print at 100% on Letter, single-sided. The content is centered at 98%; cobalt and coral bands have their own bleed layer. Select the matching paper profile and rear feeder for Epson Letter borderless printing.

Build the positioner

Build one swivel and one horizontal pivot, each with an inline NEMA 17 motor and PGFUN TYXC-0062_17_50 reducer. The gun sits in two lined housing bands supported from below. The existing rotator provides tube rotation.

The design, assembly STEP and print pack govern dimensions. All dimensions here are millimetres. The purchase and fastener allocation covers one complete fixture, controller and one camera stand.

Print and prepare

Print the journals and housings in their specified orientation. Use a hardened 0.6 mm nozzle, 0.20 mm first layer, 0.24 mm normal layers, six walls and 100% infill for the structural PET-CF17 pieces. Start from the printer's established Fiberon PET-CF17 profile within Polymaker's 270-300 C nozzle and 70-80 C bed ranges; fan off. Dry wet stock at 100 C for 10 hours and feed from the dryer. These parts use the as-printed material; annealing is not specified. Manufacturer annealed strength numbers do not establish these parts' strength.

The STL files already have their print orientation and bed Z=0. Do not use automatic reorientation. Use accessible supports on every part marked in the manifest: fork, cheeks, motor pedestal, yaw bearing cartridge, both journals, cradle/retainers, bench toes, all four switch holders, controller case, cable post sections/saddle and camera risers/upright. Support the elevated plates, flange/key projections, band cavities, front cradle shoulders, switch-clip roofs and controller cable-window roofs. Support the toes' 12 mm nut-loading lanes from their open sides. Keep support contacts out of bearing bores, screw tunnels and the short internal nut-seat shoulders; these bridge. The camera-post bores have 14 mm closing bridges; the lens bolt wells have 2.6 mm radial shoulders. Inspect those bridge toolpaths in the slicer and their printed seats before installing hardware. Use manual support painting and blockers to keep removal paths open. The cable-post service bores open at both ends for inspection and cleanup. Remove all support material without rounding seats or changing housing contact profiles. PETG camera/control pieces use the established PETG profile, four walls and 25% infill; the thin walls remain solid. TPU liners use three walls and 100% infill. The 0.2 mm lens pad is one first layer.

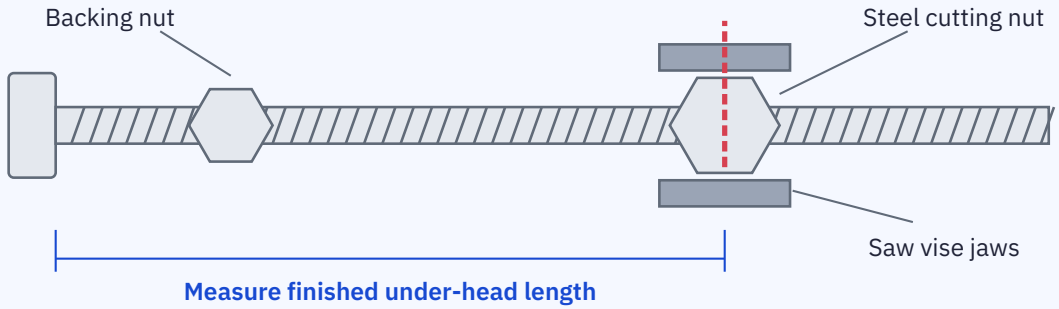
Fit the bearings to the actual printed journals and housings. A journal enters with firm hand pressure and cannot turn in the inner ring under hand torque. The housing accepts the outer ring squarely by hand without rocking. Adjust the corresponding dimension in `design.json` and regenerate the affected part if its fit fails. Do not force a bearing through its balls, sand a loose seat larger, or glue over a rocking fit.

The nominal spacers are 7 mm for the paired 6808 bearings. Print inner and outer spacers together and compare their heights with the owned caliper; keep their difference within 0.02 mm. After assembly both races must turn smoothly without axial rattle. Trim a high spacer squarely or regenerate a matched pair; do not tighten caps until a binding stack frees up.

Load all captive nuts, including the M6 follower nuts, before closing a joint. Nut loading windows open to the outside on the bearing cartridges, opposite bearing cap, cradle rails, frame tie and cable post. The output journal flange nuts load from its underside before the journal enters its bearing stack.

Build the positioner

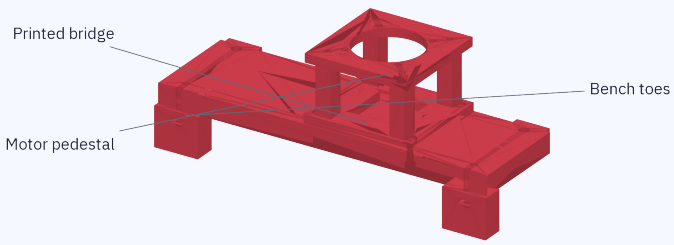
SHORTEN M3 STOCK / SCHEMATIC



Cut through the sacrificial nut and screw; deburr, then back the other nut over the end.

Cut two purchased M8x80 anchors to **45 +/-0.5 mm** under-head length. Cut owned M5x50 screws into eight 20 mm, four 25 mm and four 30 mm screws. Four owned M3x60 screws supply 30 mm fan bolts; two more supply 40 mm pitch switch-holder bolts. Thread a backing nut onto each screw before cutting; deburr, back the nut over the cut and confirm another nut runs freely. For the M3 cuts, clamp a sacrificial steel nut around the cut location in the metal saw's vise and cut the nut and screw together: the nut supplies the section for the owned 24 TPI blade. Support the work in the vise; do not hold a small bare screw at the blade. Measure every finished length. Use ordinary hex nuts as the captive fasteners.

1. Mount the bridge



Coral shows the bridge, bench toes and pedestal added to the stationary rotator base.

AT THIS OPERATION

`bridge`, two `bench-toe` feet, two M8x45 anchors with washers/nuts, four M5x30 toe bolts, four M5x25 pedestal bolts and `motor-pedestal`.

01 With the tube removed, put the bridge over the two near-side stationary base holes, X=-102 and X=162, Y=-107. The 9.7 mm locating collars enter the existing 10 mm holes. The bridge seats on the 12 mm base.

02 The two toes support the bridge's outboard edge on the bench. Install their M5 nuts through the side windows and join each toe with two M5x30 bolts. The feet bottom at the same bench plane as the rotator's feet.

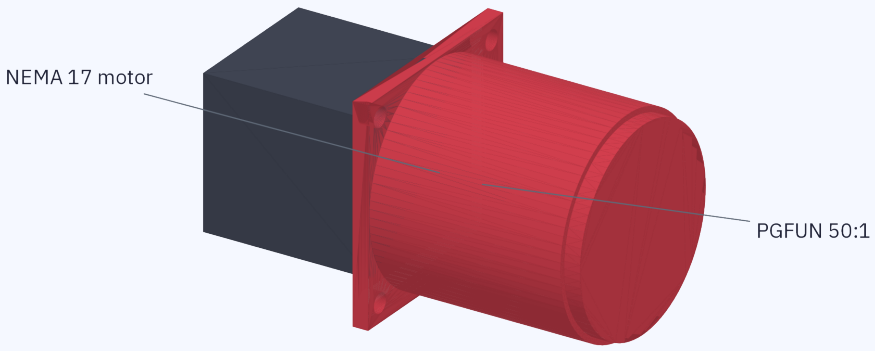
03 Insert the M8 anchors through bridge and base. Their shortened ends clear the bench. Snug washers and nuts until the bridge cannot rock; do not bow the existing base. Preserve the rotator's existing bench attachment.

04 Load four M5 nuts into the bridge underside, then attach the pedestal with four M5x25 bolts through its recessed head seats. Their ends must stay clear of the bench and moving rotator.

READY CHECK

every foot bears on the bench, the bridge has no rocking motion, and turning the empty rotator causes no contact. A rocking bridge requires flat mating surfaces and corrected foot height before fastening.

2. Assemble the motor drives



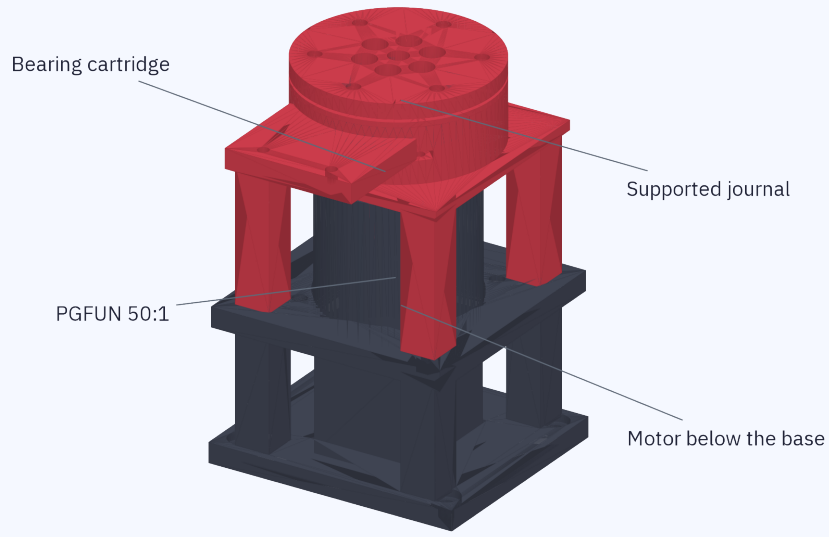
Each drive uses one NEMA 17 motor and one PGFUN 50:1 reducer. Remove the input flange to reach the motor screws.

Follow PGFUN's **"Harmonic Reducer Assemble Instruction (Nema 17)"** video on the selected Prime listing. Its input flange is removable to expose the motor mounting screws. Preserve its seal and supplied case screws; the outer four 5.5 mm mounting holes do not retain the motor. The video also covers the input-shaft side set screws.

Remove that flange, attach it to the motor's 31x31 mm threaded pattern from inside using four M3x8 screws per motor, retain the seal, seat the 22 mm pilot and 5 mm motor shaft, reinstall the flange and secure the input coupling as the supplier demonstrates. Confirm that the actual flange leaves **3-4 mm of M3 thread engagement**, the screws do not bottom in the motor, and the shaft is clamped on its flat. Use the supplied motor screws if their length is specified differently. Turn the motor shaft by hand before installing the units; the output should move smoothly through reduction.

The model uses the supplier's 60 mm square input flange, 70 mm outer bolt circle, 63.1 mm overall length and six M4 output holes on a 30 mm circle. Verify those interface dimensions against the received units before fastening printed parts. An incoming dimensional discrepancy requires a corrected adapter; it is not addressed by enlarging the bearing seats.

3. Install the swivel



The supported journal and cartridge carry the fixture load. Align the reducer case after the bearing stack is seated.

AT THIS OPERATION

yaw motor/reducer, `yaw-output-adapter`, `yaw-journal`, two 6808 bearings, inner/outer spacers, `yaw-bearing-cartridge`, `yaw-bearing-cap`.

01 Fasten the reducer's square input flange to the pedestal with four M5x20 bolts, washers and nuts. Leave them finger snug. The motor hangs below the reducer, with its cable exiting through the pedestal opening.

02 Load six M3 nuts into the output adapter. Attach it to the output with six M4x10 bolts and flat washers in the deep head wells. Effective M4 engagement is about 5.2 mm in the vendor's 10 mm deep holes. Confirm neither screw tips nor heads bottom; keep washers within the 9.4 mm wells.

03 Load the six M5 flange nuts into the journal underside. Join journal to adapter with six M3x12 bolts and owned 7 mm OD x 0.5 mm washers. The six keys enter their matching pockets; they carry torque. Seat the faces squarely without crushing the keys. The key sides have 0.05 mm nominal interference, which the assembly clearance receipt identifies explicitly.

04 Fit the bearing cartridge around the journal. Stack: first 6808, 7 mm inner and outer spacers, second 6808. Seat outer rings on the cartridge shoulder and inner rings on the adapter. Support the ring being pressed.

3. Install the swivel

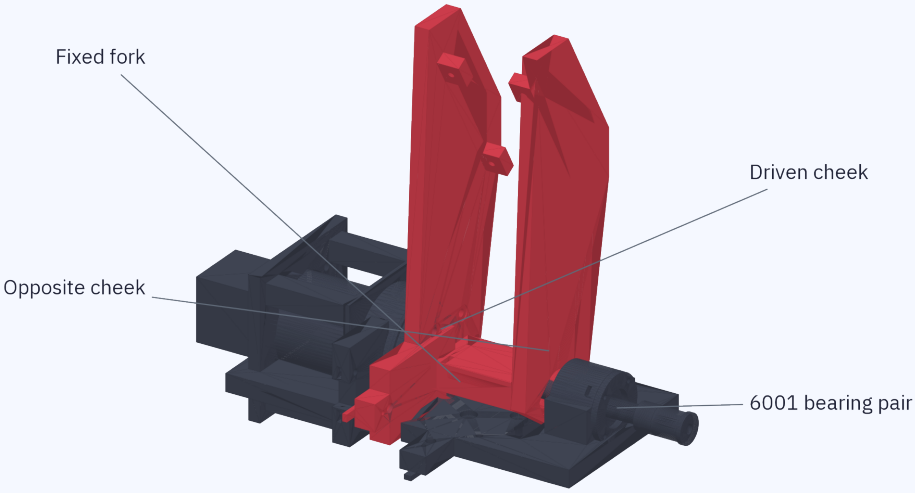
05 Attach the cap with four M4x16 screws through its outer loading-window nuts. Snug evenly; it captures the outer races with 0.1 mm nominal axial clearance. Journal and output flange must turn without rubbing it.

06 Fasten cartridge legs to the pedestal with four M5x20 bolts. Turn the assembly by hand with the motor unpowered, centre the floating reducer mounting holes, then tighten its case bolts evenly. The external bearings carry the gun frame; the reducer delivers torque through the adapter.

READY CHECK

no axial rattle, no tight spot and no cartridge movement. Binding calls for coaxial alignment or matched spacer/seat dimensions.

4. Install the horizontal pivot



Coral shows the driven and opposite cheeks and the bottom tie. Both sides support the cradle.

AT THIS OPERATION

`fork`, pitch motor/reducer and supported journal stack, left/right cheeks, `bottom-frame-tie`, two 6001 bearings, 12x100 mm axle, five axle spacers, opposite bearing cap and keeper.

01 Assemble the cheeks and bottom frame tie first. Four M4x20 bolts run through the cheek recesses into the tie's side-loaded nuts. These bolts become inaccessible behind the output flange after installation.

02 Assemble the pitch reducer, adapter and paired 6808 journal stack by the same sequence as the yaw stack. The output points toward the left cheek. Its case mounts to the fork with four M5x20 bolts and nuts. Leave the case bolts loose for coaxial alignment.

03 Attach the left cheek to its journal flange with six M5x16 bolts. Insert the axle from the right through the split collet in the right cheek; its inboard end is at pivot-local X=30.

04 The axle stack, proceeding outward, is: 8 mm collet spacer, first 6001 (8 mm), 8 mm inner spacer, second 6001 (8 mm), 21 mm front stop spacer, 12 mm middle spacer, 18 mm end spacer, 3 mm keeper. The 6001 outer rings sit in the fork at X=55 and X=71. The keeper starts at X=130.

4. Install the horizontal pivot

05 Load four M3 nuts through the opposite cartridge's radial windows and install its cap with four M3x12 bolts. The cap captures outer races. Secure the keeper with the shaft supplier's matching M6 end screw; confirm engagement and that it does not bottom in the tapped end.

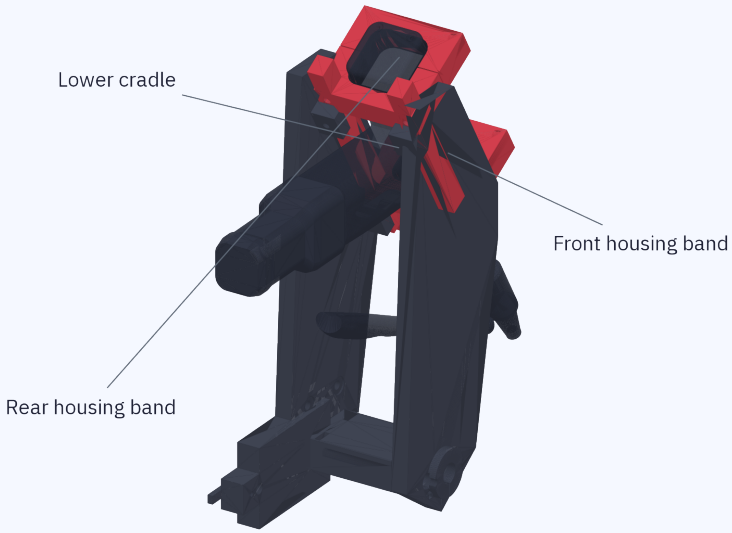
06 Clamp the right cheek collet with one M4x20 bolt. Tighten only enough to prevent shaft slip. The spacing stack must seat without forcing the bearing rings sideways or pinching them.

07 Centre the pitch reducer on the external bearing axis, then tighten its case bolts. Fasten the fork to the yaw journal's six flange nuts with six M5x16 bolts and washers.

READY CHECK

the assembled frame turns smoothly at both axes, the shaft cannot slide outward and the two cheeks stay parallel. Rotate only through the stop-defined range after installing stops. The gearmotor is not used as a press or as a means of forcing a crooked bearing stack straight.

5. Secure the gun from below



Two lined housing bands retain the gun. The lower cradle carries its weight; the forward shoulder stops axial sliding.

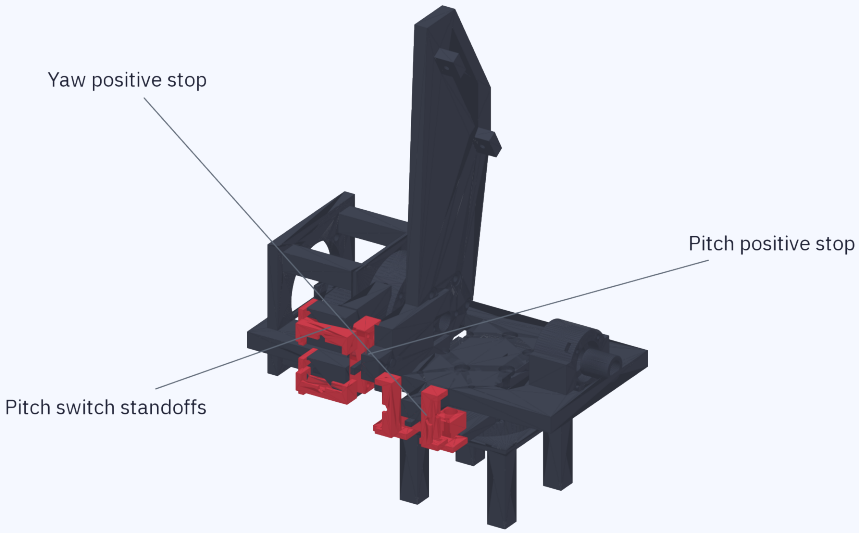
AT THIS OPERATION

`lower-cradle`, two gun retainers, four TPU band liners and eight M4x20 bolts/nuts. Keep the welder powered off during fitting.

Attach the lower cradle to the four integral cheek pads with four M4x20 bolts into its rail nuts. Put the lower TPU liners in the two bands, lay the measured scan housing into them, and put the upper liners and retainers over the housing. Each retainer uses two M4x20 screws into the lower band's captive nuts. Tighten alternately to compress the liners slightly and prevent housing slip; the nominal 0.6 mm split permits compression. Do not bottom the split or crush the housing. The front shoulder wings provide an axial stop while clearing the wire-feed guide. The grip, mode button, factory trigger and boot remain accessible.

The supplied print pose is 60 degrees below horizontal and 16 mm from nozzle tip to nominal joint. Those are editable mounting parameters, not a recorded welding recipe. Establish the actual focus and working attitude on a sacrificial joint using the X1 Pro focus procedure. If the printed pose misses that setting, change the cradle/cheek parameters and regenerate the parts before a welding run; motor trim is reserved for the small in-process correction range.

6. Set stops and switches



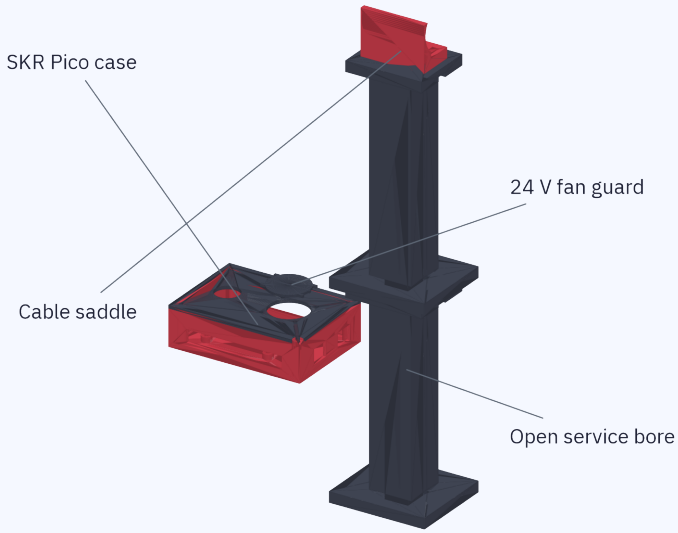
Coral marks the positive-stop sectors and switch holders. Set switch opening before either mechanical stop.

Attach each fixed stop sector with two M4x16 bolts. Put an M6x25 follower through the moving fork and an M6x35 follower through the left cheek's integral stop arm, at the 70 mm radius. The pitch sector mounts on the left-side fork ear. Its thread crosses the sector slot. The nominal hard stop is +/-1.5 degrees; soft travel is +/-1 degree. Confirm actual assembled hard-stop travel is at most 1.7 degrees and nothing else touches first.

The two M3x25 parking screws pass through the matching 3.05 mm datum holes at radius 77.5. Insert with motor power off. Bias each axis consistently toward the same side of its pin clearance, then remove both screws before referencing the controller. They establish a repeatable assembly datum; the camera qualification establishes how repeatable that datum actually is.

Four KW12 roller switches sit in the printed holders. Use M3x12 for yaw holder mounting. Mount the pitch holders through their two 30 mm standoffs with owned M3x60 screws shortened to 40 mm. Use M3x8 side screws to grip each switch's body, clear of its lever. Slide the body and holder until the moving cam opens the normally closed contact at 1.15-1.35 degrees in each direction. Show that the NC contact opens before the follower reaches its hard stop. Adjust one switch at a time with power off, then electrically check all four. Wire the two NC contacts per axis in series, as in control.md.

7. Support cables and controller

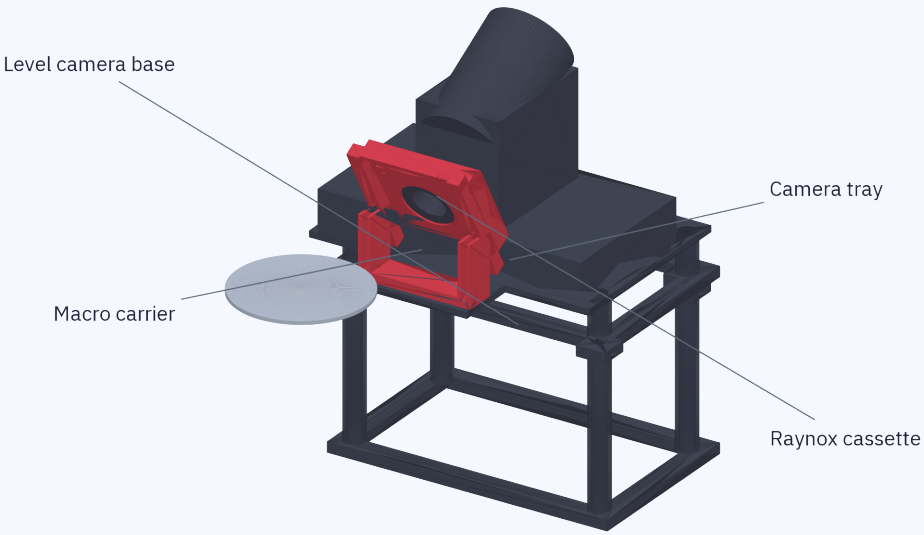


The cable post unloads the gun-side cable. Its central bore remains open for inspection and support removal.

Join the two printed cable post sections with two M5x30 bolts; mount the saddle with two M5x20 bolts. Use the four owned 25 mm fender washers at these accessible top joints. Clamp the post foot to the bench using two owned C-clamps. Route the existing loose cable curve over the broad saddle and tie only its jacket. Preserve its natural bend and slack; do not force a tighter fiber bend or let a tie flatten the cable. The gun-side residual cable force must satisfy the commissioning load limit.

Mount the SKR Pico on four M3x10 screws and 7x0.5 mm washers, with captive nuts loaded from the case bottom. Install the 22 mm NC stop button in the lid. Attach the 40x40x20 mm 24 V fan and printed guard using four M3x30 screws, washers and nuts. Fit the lid with four M3x12 screws. Cable exits keep the USB, motor, stop and 24 V harnesses clear; secure their strain relief with owned ties. Keep the controller away from the weld plume.

8. Set up the camera



The camera base stays level. The head and macro cassette look down at 29 degrees during dry observation.

Join `camera-riser-base` and `camera-riser-top` with four M5x20 screws. The tray is horizontal, as required by the camera manufacturer. Its PTZ head and the separate macro cassette look down at 29 degrees, within the head's 30-degree downward limit. The nominal sight line clears the 6.35 mm recessed rim by about 1.6 mm. Secure the camera base through both pairs of tray tie slots using the owned 8-inch ties chained into long straps. Grip the camera base, clear of moving PTZ housing, connectors and ventilation.

Fasten the lens upright to the tray with two M5x20 bolts/nuts. Install the Raynox DCR-250 in the cassette: rear 43 mm neck enters the shoulder, outer front rim contacts the 0.2 mm TPU pad, then the retainer holds the rim with four owned M3x25 screws. Keep the optical aperture and glass clear; the clamp bears only on the metal body. The optional 0.4 mm TPU shim takes up received axial clearance if needed. Do not use the universal snap-on adapter inside this cassette.

Attach the cassette behind the upright's two long slots with four M5x30 screws and nuts. Insert screws from the cassette's rear head wells; the carrier lies on its front face, outside the optical aperture. Align its centre to the camera's actual optical axis, set the camera's lens to face through it, and slide the whole stand to focus near Raynox's approximately 109 mm working distance. The published camera body envelope governs tray clearance; the optical-centre illustration is not a measured camera mounting datum. Slots provide +/-49 mm adjustment along the tilted lens plane. Set the head and cassette coaxially using those slots and stand position.

8. Set up the camera

Before any camera power-up or reboot, remove the lens upright and cassette as a unit using its two foot bolts. The camera's automatic pan/zoom self-test needs that space. Reattach only after self-test, with the head set to its recorded pose. Never turn its motorized head by hand.

Lock focus, exposure and PTZ/tracking for the dry measurements. Keep the actual wire endpoint and aiming dot in the same sharp view. Withdraw this entire stand before emission: this build uses learned trajectory replay and supplies no qualified welding-light protection for the camera.

Fastening and release check

Use flat washers under bolt heads against prints. M3 wells take the owned 7x0.5 mm washers; M4/M5 wells take the kit's matching flat washers. Use one owned M5 hex nut in addition to the kit's 45 nuts. Start printed joints at M3 0.15, M4 0.3 and M5 0.5 N m, then use the lowest preload that passes retention and movement checks. These are assembly starting points, not qualified joint capacities. Do not apply these values to the reducer's factory screws: use its installation instructions. Use Loctite 243 only on compatible metal-to-metal threads and allow its stated cure; keep it off plastic and bearings. Add witness marks to motor couplings, journal adapters, shaft clamps and base anchors.

Complete commissioning before running with the tube present. Its procedures address loaded retention, stop travel, heat drift, camera noise and independent dry laps. Record received fit and real results beside the hardware; numerical clearance is not a load or lifetime result.

Parts and fasteners

All linked listings were observed with Prime in the signed-in Amazon buy box on 2026-10-05. Prices below are full purchase packs before tax; delivery observations are estimates at that time. Select PGFUN **50:1** and FoMaKo **Classic Black K20UH** explicitly. The gearmotor combines separate motor and reducer purchases; PGFUN does not include the motor.

Item	Buy	Price each	Total	Prime delivery observed
PGFUN TYXC-0062_17_50 harmonic reducer	2	$139.99	$279.98	Oct 7
STEPPERONLINE 17HS19-2004S1 motor with cable	2	$14.99	$29.98	Oct 6
SKR Pico V1.0, integrated TMC2209	1	$35.99	$35.99	Oct 6
BTF 24 V 4 A adapter with barrel breakout	1	$26.99	$26.99	Oct 6
HARFINGTON 6808 bearings, 40x52x7, four-pack	1	$12.40	$12.40	Oct 6
uxcell 6001-2RS, 12x28x8, two-pack	1	$5.99	$5.99	Oct 6
12x100 mm M6-end shafts/bolts, two-pack	1	$12.99	$12.99	Oct 6
24 V 40x40x20 fans, two-pack	1	$12.99	$12.99	Oct 6
Fgruh 1220-piece M3/M4/M5/M6 assortment	1	$22.99	$22.99	Oct 6
M8x80 bolt/nut/washer sets, four-pack	1	$9.59	$9.59	Oct 7
22 mm latching NC stop button	1	$13.99	$13.99	Oct 6
KW12-3 NC roller switches, ten-pack	1	$5.99	$5.99	Oct 6
Loctite 243	1	$9.49	$9.49	Oct 6
Mechanism and controller			**$479.36**	
FoMaKo K20UH Classic Black	1	$449.00	$449.00	Oct 6
Raynox DCR-250	1	$75.50	$75.50	Oct 6
Camera and lens			**$524.50**	
Complete new purchase subtotal			**$1,003.86**	

Parts and fasteners

The four 6808 bearings, both 6001 bearings, one shaft and one fan are used. The shaft's supplied matching M6 end screw retains its keeper. The camera includes its 12 V adapter and USB3 A-to-B cable; the acquired Anker hub supplies the Mac's USB-A connection. General tools, rotator, welder, host, two C-clamps, resistors, wiring/ferrules/WAGO stock and the listed reused fasteners are on hand. No Raspberry Pi, metal camera stages or extra laser-interface adapter are required by this build. Purchases have not been placed by this release.

Exact screw allocation

Screw	Demand	Kit supply used	Reused/prepared supply	Assignment
M3x8	12	12 of 35	0	8 motor flanges, 4 switch body clamps
M3x10	4	4 of 35	0	Controller board
M3x12	22	22 of 35	0	12 adapter/journal, 4 opposite bearing cap, 2 yaw switch holders, 4 case lid
M3x25	6	0	6 owned	4 lens retainer, 2 removable parking pins
M3x30	4	0	4 owned M3x60 cut to 30	Fan and guard
M3x40	2	0	2 owned M3x60 cut to 40	Pitch switch holders through 30 mm standoffs
M4x10	12	12 of 25	0	Six reducer-output screws per joint
M4x16	12	12 of 25	0	Eight 6808 caps, four stop-sector mounts
M4x20	13	13 of 25	0	4 retainers, 4 cradle pads, 4 frame tie, 1 right axle collet
M5x16	12	12 of 12	0	Six yaw fork and six driven cheek flange screws
M5x20	20	12 of 12	8 owned M5x50 cut to 20	8 reducer cases, 4 yaw cartridge, 2 saddle, 4 camera riser, 2 lens-carrier feet
M5x25	4	0	4 owned M5x50 cut to 25	Pedestal to bridge
M5x30	10	6 of 6	4 owned M5x50 cut to 30	4 bench toes, 2 cable-post join, 4 macro-cassette clamps
M6x25	1	1 of 6	0	Yaw positive-stop follower
M6x35	1	1 of 6	0	Pitch positive-stop follower
M6 shaft-end	1	0	Supplied with shaft	Opposite axle keeper
M8x45	2	0	2 purchased M8x80 cut to 45	Stationary rotator anchors

M5 nuts demand **46**: kit 45 plus **one acquired M5 hex nut**. Use four acquired 25 mm fender washers at the cable post join/saddle; the remaining 42 M5 head washers fit the kit's 45. M4 nuts demand 25 and M3 nuts demand 36 plus six sacrificial cutting nuts; their kit supplies are 100 and 140. The two M6 follower nuts/washers come from the kit. M8 sets and shaft end hardware are supplied with their purchases. Parking pins do not use nuts.

M3 head wells use acquired 7 mm OD x 0.5 mm flat washers. Other print-facing heads use matching kit flat washers. Check motor-flange head-well clearance before adding a washer there; supplier-installed screws govern that metal interface. Spare kit screws of unlisted lengths are not counted as demand.

The acquired M5x50 40-pack supplies sixteen shortened screws. Existing M3x25 and M3x60 stock supply six each. The assembly cutting procedure uses a sacrificial nut around each small screw at the cut to suit the owned 24 TPI metal saw blade. Measure and deburr every finished under-head length.

Printed material consumption

The print manifest reports exact solid volumes, assigned materials, quantities and orientations. Structural PET-CF17 uses six walls and 100% infill. Camera/control PETG uses four walls and 25% infill. TPU liners/pads are solid. Manifest mass is **solid-equivalent**, before supports, skirts and purge; it is not a native slicer estimate. The final release receipt totals those volumes by material.

The filament is acquired stock, so its additional purchase cash is zero. Its consumed value is not zero: use the recorded PET-CF17 cost of about $39.32/kg and PETG $11.20-13.94/kg, with actual sliced/support mass, for the material charge. Budget roughly $160-200 of stock consumption for this release; replace that estimate with the native slice when printing.

Wire the controller



Use one BIGTREETECH SKR Pico V1.0 and a 24 V, 4 A supply. X operates yaw and Y operates pitch. Both integrated drivers use UART1 GPIO8/9, addresses 0/2. Preserve factory address wiring. Remove **all four DIAG jumpers** before attaching normally closed limits. Z and E0 motor outputs remain unused.

The manufacturer documentation governs connector orientation. Identify printed SIG/GND/+V on the received board; a wire-side view reverses pin order. Do not connect stop-input +V.

Harnesses

Circuit	Connection	Check
Supply	Brick centre positive to VIN+, sleeve to VIN-	Measure polarity before attachment; insulated strain relief
Yaw motor	Four leads to X motor socket	Ohmmeter identifies coils; each occupies an adjacent driver coil pair
Pitch motor	Four leads to Y motor socket	Same check; never unplug a powered motor
Yaw limits	X-STOP SIG GPIO4 -> COM/NC -> COM/NC -> X-STOP GND	Closed at centre; either end opens
Pitch limits	Y-STOP SIG GPIO3 -> COM/NC -> COM/NC -> Y-STOP GND	Same series continuity
Positioner stop	Z-STOP SIG GPIO25 -> button COM/NC -> Z-STOP GND	Latching button or broken wire opens
Rotator timing	E0-STOP SIG GPIO16 to pedal NO; E0-STOP GND to COM/rotator signal ground	3.3 V released, near 0 V pressed
Cooling	24 V fan to FAN3, positive to +, negative to switched -	GPIO20 on whenever firmware runs with VM
VM measurement	TH0 SIG GPIO27 through 8x4.7 kOhm series to VIN+; 3.3 kOhm SIG to GND	Insulated divider described below
Host	Board USB to Mac	100 ms heartbeat in motion session

The motor cable has a 0.1-inch connector. Check mating and coil order on receipt. Use acquired wire/connector stock for a secured board connection. Twist coil pairs, separate signal wiring, and tie strain relief to the case. Do not leave a loose plug carrying a moving cable's tension.

The pedal stays connected to the rotator. Its NO signal is observed in parallel. Confirm its voltage with the multimeter before joining grounds. Do not connect a 24 V pedal interface or laser work-contact loop here.

Supply divider

TH0 includes a 4.7 kOhm pull-up to 3.3 V and an ADC RC filter. With external VM-to-SIG resistance 37.6 kOhm and SIG-to-GND 3.3 kOhm:

```
V_SIG = (VM/37600 + 3.3/4700) / (1/37600 + 1/4700 + 1/3300)
```

Nominal ADC values are 2702 at 18 V, 3067 at 24 V and 3310 at 28 V. Firmware accepts 2701..3311. VM removal invalidates reference and driver verification. USB alone cannot authorize motion. The calculation includes the on-board pull-up.

Wire the controller

Insulate all resistor splices, retain the string away from bolts/fan blades, and expose only VIN/SIG/GND ends. With the board disconnected, measure the 37.6 kOhm and 3.3 kOhm sections separately. With USB and 24 V present, expect about 2.47 V at SIG. Compare status ADC against the equation using measured VM and 3.3 V. Correct a large mismatch instead of widening thresholds.

Driver settings and faults

SpreadCycle, 64 external microsteps, interpolation off, CoolStep off, current scale 13, VSENSE=0 and R110 shunts give nominal **0.774 A RMS / 1.094 A peak**. Hold current equals run current. This establishes a current setting, not measured output torque or positioning accuracy.

Fresh UART writes, IFCNT increments and register read-back precede ARM. Both drivers, stop, limits, VM, heartbeat and timer progress are monitored. A fault disables both outputs and invalidates reference. Startup/reset leave all four enables disabled. The 250 ms watchdog is fed only after timer progress.

The NC button is a firmware motion stop, not a separately rated safety relay. It does not inhibit the laser. Factory laser controls and interlocks control emission. Firmware has no laser-start command. A positioner fault stops its pulses; release the rotator pedal and factory gun trigger to stop those independent devices.

X1 control interface

The purchased isolated Waveshare RS232 adapter and DB9 breakouts suffice for receive-only capture: X1 TX pin 3 to adapter RS232 RX pin 2, ground pin 5 to 5, all other pins disconnected. The X1 manual specifies RS232; TTL and RS485 adapters cannot substitute.

DB25 pins 3/4 are enable +24 V/GND. Pins 5/6 are the factory E-stop loop. They do not define a documented emission trigger. Automatic laser control requires a captured or manufacturer-defined protocol with read-back and the factory weld sequence preserved. First welding uses the factory trigger while replaying a qualified dry trajectory. No additional interface purchase is required to begin the existing capture procedure.

Commission the fixture

These checks decide whether the assembled device can retain the gun, move usefully under load and repeat an accepted dry trajectory. Use the acquired caliper, multimeter, 2 kg x 0.1 g scale, Neoteck indicator and its rotator mount, hand hex keys, two C-clamps, witness marks and a water-filled container with a soft cord. No additional measuring-tool purchase is specified. Keep records beside this hardware, with geometry hash, material/profile, date and actual measurements. A successful check establishes only its stated scope.

Incoming and power-off checks

01 Check both reducers against the received-interface dimensions in assembly.md. Check the assembled bearing fits and matched spacers. Free rotation must have no binding or axial rattle before case alignment is locked. Record the chosen journal/bore dimensions.

02 Attach the bridge and toes. Check every foot bears on the bench and the anchors cannot rock. Retain the existing rotator's bench attachment.

03 Install the gun with four-foot cable length supported by the cable post. Put witness marks on gun/bands, keys/flanges, shaft clamp and base.

04 With motor and welder power off, sweep both axes to all four corners of their hard-stop range, while turning the rotator through one revolution. Check the entire cable, actual switch rollers, bolt heads, wire guide, recessed tube rim and controller harness. No object may contact before its intended stop; the gun/guide must clear the tube at every corner.

05 Measure follower centre displacement at 70 mm radius with the caliper. An assembled +/-1.5 degree stop gives about +/-1.83 mm. Reject travel beyond +/-2.08 mm (1.7 degrees), any earlier interference or a loose stop. With a continuity meter, each limit must open between 1.40 and 1.65 mm follower displacement (1.15..1.35 degrees), before its hard stop.

These measurements change the release decision: an incorrect stop or received interface can defeat the known clearance envelope. Correct it before power. Do not enlarge bearing seats to accommodate an unrelated mounting error.

Retention and cable load

Fill and weigh the container plus cord to **1.020 kg**, giving 10.00 N. Keep it close to the bench over a padded catch. The container is a known static load, not an impact test or an inferred hand-pull force.

With the tube removed, motors disabled and the frame supported by a soft catch, apply this load through a broad soft loop around the printed cradle, not the nozzle, trigger, fiber boot or wire guide. For a vertical load, hang it directly; for another direction, turn the detached fixture and clamp its flat bridge to present that direction vertically. Keep the force line within 240 mm of the pitch axis. The conservative gun-gravity bound plus this added load is covered by the 7 N m normal structural screen. Hold for 60 seconds, unload and repeat three times. Check band slip, base motion, key seating, shaft retention, cracks and bearing-seat movement. No witness-mark shift, permanent set, cracking or newly detectable rattle is acceptable. Do not increase motor

current to force a defective joint through this check.

The 10 N cable term is a design allowance, not an existing measured cable force. Route a relaxed gun-side curve over the post without loading the boot. The loaded camera checks below qualify that actual route. If the motor stalls, a witness mark shifts or alignment changes when the cable moves, improve support/slack and repeat the affected checks. Do not infer an independently measured cable-force bound from the retention weight test.

With the gun installed, anchor an indicator to the stationary base and touch a printed cradle face. Repeated load/unload cycles must return within one indicator division and preserve witness marks. The owned indicator is a coarse permanent-set check; its resolution does not certify 0.010 mm camera performance. Use camera observations for the loaded precision checks below.

Electrical commissioning

Use control.md. Before power, verify coil pairs, supply polarity, DIAG removal, the three closed NC loops, all six opened-loop conditions and the insulated VM divider. Test one disconnected signal wire per loop. Verify the pedal's released/pressed 3.3 V/0 V levels. No stop-input +V is connected.

Flash the supplied UF2 with motors unplugged, then remove the boot jumper. USB-only status must show disabled outputs, invalid reference and unhealthy VM. After measuring 24 V and divider SIG, power down, attach motors and power up. Status must report the released matching geometry hash, 640,000 counts/rev, current scales [13,13], VSENSE [0,0] and 64 microsteps. Both UART samples must be valid; clearing a fault does not establish a reference.

Align parking pins with power off, bias both toward the same designated side, remove them, then use `clear`, `reference central`, `arm` in the host console. Start with +/-32-count yaw and pitch jogs, 500 ms duration. Each positive command must move as the CAD model specifies. If a motor rotates oppositely, power down and reverse one complete coil pair, then repeat. If coil order or motion is uncertain, stop before increasing the jog.

Test stop button, each of four limit switches, loss of VM and host heartbeat one at a time during a small jog, away from the tube. Motion must cease, outputs disable and reference become invalid. Unplug USB or terminate the host to test heartbeat: it must fault within 500 ms plus one 250 us tick. A GPIO-open stop/limit is sampled every 250 us. These are programmed bounds; record actual behaviour. Repeat recovery from the physical parking datum for every fault. No reset, CLEAR or reconnect may resume a previous motion.

Loaded movement and heat

Run the camera procedure in observation.md, using the real gun and cable, after the controller has held the working pose for 30 minutes. Collect the motor response in all four approach-direction combinations and collect a separate holdout session. Acceptance requires each feature's uncertainty <=0.0025 mm, rank-two response with condition number <=30, 95th-percentile error <=0.005 mm and maximum <=0.010 mm. Counts that do not produce useful movement remain part of the evidence; do not relabel a count as measured travel.

Record stationary feature positions for ten minutes after warm-up, then repeat the response holdout. No witness-mark movement or bearing rattle is acceptable. Camera drift beyond 0.010 mm rejects that setup; correct heating, cable drag, exposure/focus or clamp seating before recording new holdouts. Record surface-temperature readings if an existing thermometer is available; the essential acceptance is stable loaded

observations, not an assumed material-temperature rating. Inspect fan operation throughout.

Dry runs and welding release

Index the tube physically with matching tube/nest marks. Use one direction, rotator speed and cable route. Release the pedal for at least 12 seconds before each run so its existing acceleration history matches the replay timing. Learn a periodic correction, then validate the exact candidate in **two independent dry captures**, each at least two complete revolutions. The host rejects sparse phase coverage, invalid features, stale frames, timing uncertainty, nonrepeatable runout and error outside the two-axis response. Both dot and actual wire endpoint must satisfy maximum 0.010 mm and p95 0.005 mm observed errors. A visually pleasing lap is not a substitute.

For welding, withdraw the camera stand completely, preserve the indexed setup and load only that accepted trajectory. Establish focus/attitude and a working factory weld recipe on a sacrificial joint of the intended material and thickness; neither 60 degrees nor 16 mm is a qualified process recipe. The existing factory trigger controls emission. Pedal timing controls motion replay. Begin with a short bead, inspect fusion and wire placement, then progress to the continuous circular weld after both geometry and process behaviour agree. A changed wire-feed setting, focus, tube joint, direction, speed, static pose or cable route requires new affected dry validation.

This positioner does not qualify a pressure-containing weld. The project's existing weld inspection and pressure-vessel acceptance requirements still govern the final carbonator. Record weld/process results separately from positioning results; do not infer fusion, penetration or pressure integrity from a camera alignment result.

Learn with one camera

OBSERVE
Actual dot, wire and seam

→

LEARN
Loaded motion + reversals

→

VALIDATE
Two independent dry runs

→

REPLAY
Indexed rotator + factory trigger

One camera measures projected error during dry learning. Withdraw the stand before emission.

One FoMaKo K20UH and Raynox DCR-250 view the actual aiming dot, actual wire endpoint and seam. The camera measures projected image-plane positions. It does not observe hidden depth or independently establish focus. All learning is dry; the stand withdraws before welding. The qualified path then replays continuously from the existing rotator pedal.

Use the mechanical and electrical checks first. The following commands run from the repository root with `tools/cad-venv/bin/python`. Files filled with observations/configurations belong outside this public repository. Examples use `/tmp/pgfun`; use a durable personal folder for accepted sessions. Replace the camera ID and serial-device examples with actual enumerated values.

Camera setup

Keep its base level. Withdraw the macro upright/cassette before every camera boot: automatic self-test moves the head and zoom. After self-test, set the head to the recorded 29-degree downward view and reattach the macro carrier. Use the supplied 12 V adapter and USB3 A-to-B cable through the acquired Anker 332 USB-C hub. Camera and positioner use its two USB-A data ports; the unrelated panel camera is disconnected during these measurements.

Set the camera's HDMI video format to 4K and enable USB 4K mode. The camera manual makes these prerequisites for native USB 3840x2160 output; its LAN/NDI stream becomes 1080p in that mode. The helper and new observer require native 3840x2160 frames. A scaled 1080p stream does not satisfy this procedure.

With the camera's web/OSD controls, set manual focus, manual exposure, fixed white balance and auto-tracking off. Record zoom, focus, exposure and pan/tilt settings in the local configuration. The new observer's booleans are operator records, not a camera read-back. Any setting or stand movement requires new scale/noise calibration and affected learning.

Start near the DCR-250's approximately 109 mm working distance. Adjust stand, head, cassette and optical zoom until the actual wire endpoint, red dot and seam are simultaneously sharp and visible across the intended small sweep. Start with about 10 mm field width. Reject clipped dot pixels, reflections that compete with the dot, and a wire silhouette merged with the seam. Reduce exposure or improve lighting/angle instead of identifying the wire nozzle as the endpoint. Check rim occlusion in the received setup.

Learn with one camera

Scale, regions and stationary noise

Enumerate the physical camera and copy the template:

```
mkdir -p /tmp/pgfun
tools/cad-venv/bin/python firmware/src_pgfun_positioner/host/observe.py --list
cp firmware/src_pgfun_positioner/host/optics.example.json /tmp/pgfun/optics-initial.json
```

Fill `dot`, `wire` and `seam` ROIs as `[x,y,width,height]` in native full-frame pixels. Each should isolate its actual feature. Set wire approach direction and dark/bright polarity from the real image; configure seam orientation from its local straight segment. ROIs are arrays, not dictionaries.

Calibrate tangent and normal scales separately in the joint's reference plane, using a known long dimension measured with the acquired caliper. A 20 mm baseline reduces the relative effect of its coarse resolution. Repeat at an independent length/location and record the scale discrepancy. For a small field, acquire the long baseline at a wider recorded zoom, then calibrate the final zoom directly with the largest known dimension that fits; do not assume a zoom-ratio conversion. A temporary coupon with caliper-checked edges in that plane supplies the reference. Do not use commanded motor counts as the length reference. Include scale-error contribution over the observed correction range in the 0.0025 mm measurement uncertainty allowance.

Set `mm_per_pixel` to the two measured values, `manual_focus` and `manual_exposure` true, `auto_tracking` false, and an initial conservative `stationary_sigma_px` of 1. Keep `noise_qualified` false. Save a native camera frame using its camera application or the shared helper's frame-export path, and retain it with ROI/scale records. Focus and exposure must remain fixed.

Record a 30-second stationary sequence with motors holding the working mass and cable after warm-up. The observer can collect this initial sequence; the motor-response learner refuses it until noise is qualified.

```
tools/cad-venv/bin/python firmware/src_pgfun_positioner/host/observe.py \
  --camera-id 'ACTUAL_CAMERA_ID' --config /tmp/pgfun/optics-initial.json \
  --latest /tmp/pgfun/latest.json --log /tmp/pgfun/stationary.jsonl --seconds 30
tools/cad-venv/bin/python firmware/src_pgfun_positioner/host/noise.py \
  /tmp/pgfun/stationary.jsonl /tmp/pgfun/optics-initial.json \
  /tmp/pgfun/optics.json --receipt /tmp/pgfun/noise-receipt.json
```

The noise check requires 300 valid physical frames, at least 15 seconds, 98% valid coverage, sigma <=0.0025 mm, p95 <=0.005 mm and maximum <=0.010 mm. It derives an empirical pixel-jitter floor. The live observer also propagates feature and seam-angle uncertainty; stationary jitter alone cannot certify absolute endpoint accuracy or scale bias. Improve optics/support if the result fails, without lowering the requirement or inventing a noise value.

Restart the observer with `/tmp/pgfun/optics.json`, the same camera ID, latest path and a new log, with `--seconds 3600`. Keep that process running in a separate terminal for all dry learning. Frames older than 200 ms, sequence gaps, clipped dots and invalid features cannot control motion.

Loaded responses, including reversals

Align and remove both parking pins before **each** motion session. The `--reference-central` option records that physical action; it does not perform it. Use the same pin bias, tube setup, temperature and cable route.

Learn with one camera

```
tools/cad-venv/bin/python firmware/src_pgfun_positioner/host/servo.py learn \
  --port /dev/cu.YOUR_POSITIONER --latest /tmp/pgfun/latest.json \
  --output /tmp/pgfun/train.json --reference-central
tools/cad-venv/bin/python firmware/src_pgfun_positioner/host/servo.py fit \
  /tmp/pgfun/train.json /tmp/pgfun/model-candidate.json
```

Repeat `learn` into `/tmp/pgfun/response-holdout.json` after a fresh physical datum, then qualify:

```
tools/cad-venv/bin/python firmware/src_pgfun_positioner/host/servo.py qualify \
  /tmp/pgfun/model-candidate.json /tmp/pgfun/response-holdout.json \
  /tmp/pgfun/model.json
```

The learner probes 2/4/8/16/32/64-count levels with two independent vectors in each of four direction branches. Each probe returns from the opposite approach before observation. It fits a loaded 4x2 response and per-axis reversal deadband of 0..32 counts. Deadband at the search boundary calls for larger bounded probe levels or mechanical correction; it is not established as <=32 simply because the search stops there. The holdout requires five independent responses per branch and the stated error limits. Inspect all fitted deadbands and useful movement before accepting the model.

At a focused, physically correct working pose, copy a fresh valid `latest.json` to `/tmp/pgfun/target.json`. Check its dot/wire positions and retain a frame showing both at the intended joint. The target must have the same optical-configuration digest as the response model. Do not use a synthetic coordinate or the commanded count-zero as the target.

Learn periodic correction

Index the tube/nest physically, choose one rotator direction and a speed of 5..15 mm/s, and release the pedal for at least 12 seconds before each run. The example uses **8 mm/s, clockwise, index label** `tube-zero`. This gives about 48.6 seconds per revolution. Keep camera observation running.

```
tools/cad-venv/bin/python firmware/src_pgfun_positioner/host/trajectory.py capture \
  --port /dev/cu.YOUR_POSITIONER --reference-central \
  --latest /tmp/pgfun/latest.json --output /tmp/pgfun/baseline.json \
  --log /tmp/pgfun/baseline-control.jsonl --seconds 110 \
  --speed 8 --direction cw --index tube-zero
tools/cad-venv/bin/python firmware/src_pgfun_positioner/host/trajectory.py fit \
  /tmp/pgfun/baseline.json /tmp/pgfun/model.json /tmp/pgfun/target.json \
  /tmp/pgfun/path-candidate.json
```

Hold the existing pedal when the capture reports ready. The positioner waits 20 ms debounce plus 200 ms to match the rotator's cold-start settling. Capture at least two full revolutions and 512 observations, with three independent observations in every one of 64 phase bins. Host/device clock uncertainty must be <=5 ms. A three-harmonic fit learns repeatable runout; nonrepeatable residual beyond 0.005 mm rejects the setup.

The inverse includes the measured reversal branches and lost-motion allowance. That compensation is a candidate until measured during replay. It is not a claim that an advertised backlash value was removed exactly. All 256 path knots and their interpolated cubic envelope must satisfy travel, rate and acceleration limits. Required correction outside 0.250 mm local neighbourhood or outside the two-axis span calls for improved static alignment.

Learn with one camera

Independent dry validation and welding

Capture the candidate twice using the preceding capture command with `--profile /tmp/pgfun/path-candidate.json`, separate output/control-log filenames, and a fresh physical index/datum before each. Keep identical speed, direction, index label and optical settings. Then:

```
tools/cad-venv/bin/python firmware/src_pgfun_positioner/host/trajectory.py qualify \
  /tmp/pgfun/path-candidate.json /tmp/pgfun/dry-a.json /tmp/pgfun/dry-b.json \
  --output /tmp/pgfun/path.json
```

Both independent replays must keep dot and actual wire errors at maximum 0.010 mm and p95 0.005 mm. If a repeatable residual remains, fit a revised candidate using a captured replay and the same qualified model/target, then collect two new holdouts of that exact candidate. Training and validation samples may not overlap. Repositioning or changed optics require a new model.

After an accepted dry result, stop observation and withdraw the entire camera stand. Match the accepted tube index, datum, cable route, speed and direction. Use the factory gun trigger with the working weld recipe while replaying:

```
tools/cad-venv/bin/python firmware/src_pgfun_positioner/host/trajectory.py replay \
  --port /dev/cu.YOUR_POSITIONER --reference-central \
  --profile /tmp/pgfun/path.json --seconds 60 --log /tmp/pgfun/weld-control.jsonl
```

Release the pedal to stop the lap and factory trigger to stop emission. Replay has no camera input and no rotator angle encoder; its qualification is specific to the independent dry repeatability of that setup. It is a continuous motor path, not a sequence of stop-and-settle camera corrections. Automatic emission control depends on the separate X1 interface work.

Engineering reference

The fixture supplies a vertical swivel and a horizontal pivot. Their axes intersect 145 mm behind the nominal joint, at the joint's 208.05 mm height. The movements produce orthogonal small arcs at the aiming point. At one degree the first-order displacement is 2.53 mm; departure from the tangent is 0.0221 mm. The host uses rotations and measured camera responses.

Geometry and load paths

The bridge attaches to the stationary rotator's two existing 10 mm holes, at X=-102/162 and Y=-107. Locating collars register it; M8 anchors retain it; bench toes support its rear edge. It transfers load into the stationary base and bench, behind the moving turntable.

The yaw pedestal supports two 6808 bearings through a separate cartridge. Its journal carries the fork. The pitch journal drives the left cheek through six M5 bolts; two 6001 bearings support the right cheek's 12 mm axle. The cheeks and bottom tie support the lined gun cradle from below. Each reducer is aligned to the bearings before its case bolts are tightened. Its motor shaft is an input coupling, not a cantilever supporting the gun.

Six printed keys at 20 mm radius transmit torque from each reducer adapter to its journal. Six M3 bolts retain the faces. Keys have 6 mm depth and 0.05 mm nominal side interference. Journal flanges use six M5 bolts on a 58 mm circle. The opposite axle clamp rotates the bearing support with the frame; it is not part of the positive pitch stop.

Stops act directly between driven frames and fixed supports. An M6x25 follower in the fork stops yaw; an M6x35 follower in the left cheek's integral rear arm stops pitch. Each bears against a sector at 70 mm radius. No friction clamp on a smooth shaft is relied on to stop pitch. Four normally closed roller switches open before the mechanical stops.

Load calculation

The measured gun, guide and four-foot cable mass is **1.3118 kg**. The calculation places that entire mass at the most adverse point of the pinned exterior reconstruction. It adds gravity moments of moving prints and **10 N** residual cable force at the maximum exterior radius. This overstates the distributed gravity load; it does not assume a measured centre of gravity. The cable post must keep residual force within that allowance. Exact values and bindings are in load-screen.json.

The reducer's advertised continuous output torque is 7 N m. Normal service must stay within it. An **18 N m structural fault screen** covers nominal ideal multiplication of the selected motor's holding torque at configured peak current, with a small allowance. This is not a reducer overload rating or a calibrated electronic torque limit. A failed limit followed by a powered stall can damage the reducer; positive stops limit travel with the gun retained.

Screens cover key bearing pressure, net journal torsion, bearing moment couple, flange fasteners, stop slot pressure, stop-arm bending and an 80 mm net-width bridge section. Journal torsion subtracts the central bore and six large screw wells. The 14 mm bearing-centre spacing gives a 500 N couple at 7 N m, below the bearing seller's 4,180 N static-load figure. That rating does not qualify printed seats; their average projected contact

pressure is screened separately.

PET-CF17 calculations assume **5 MPa allowable stress** and **1,500 MPa elastic modulus**. These are engineering assumptions for as-printed parts, not measured properties. Polymaker's published strength values use parts annealed at 120 C for ten hours; this build does not claim them. Hole stress concentrations, layer adhesion, creep and contact deformation require the specified loaded checks. The screens size sections; they do not establish fatigue life or micron-scale stiffness.

Useful motion

The controller issues 640,000 counts per output revolution: 200 motor steps, 64 external microsteps and 50:1 reduction. A count at the 145 mm lever is **1.42 micrometres**. Advertised 20 arcsecond backlash corresponds to **14.1 micrometres** there. These supplier and command values do not establish received transmission error or minimum useful movement under load.

Calculated elastic twist at 7 N m is substantially larger than one count. Static compliance can enter a learned response; changing cable load, heat or slip cannot be removed by quoting backlash. Camera qualification therefore checks real dot and wire motion under working mass, routing and temperature, in all four combinations of approach directions.

Soft travel is +/-1 degree; nominal stops are +/-1.5 degrees. Firmware limits rate to 1,000 counts/s and acceleration to 4,000 counts/s squared. Maximum speed is 0.5625 output degrees/s, or 4.69 motor rpm.

The camera measures four coordinates: tangent/normal positions of the dot and actual wire endpoint relative to the seam. Two axes correct only their rank-two span. Residual error outside that span rejects the trajectory. Correct the nominal cradle pose or wire-guide relationship instead. One camera cannot determine invisible depth or welding focus.

Verification scope

clearance-check.json screens full-precision fixture meshes against pinned gun and rotator CAD at 25 combinations spanning both hard-stop ranges. Intentional key interference and stop contacts are recorded separately. Real cable shape, switch rollers, fastener heads and continuous swept volumes require the power-off assembly sweep. Each printable STL is a closed positive volume within the H2C envelope.

Individual STEP, STL and viewer payload share the print frame. Assembly CAD uses rotator coordinates. Controller and host profiles bind to the parameter hash. A parameter change requires new CAD, firmware and affected physical checks. Commissioning specifies fit, retention, interlocks, loaded motion and thermal drift. Observation specifies camera setup, learning, independent dry validation and replay. Physical results are recorded only after assembly and measurement.

Manufacturer references

- PGFUN reducer and installation video: TYXC-0062_17_50 drawing, torque and backlash claims.
- BIGTREETECH SKR Pico: schematic, board drawing and pin assignments.
- PET-CF17 technical data: drying, printing and annealed test conditions.
- Raynox drawing: lens body, neck and axial dimensions.
- X1 Pro manual: focus and external interfaces.

Firmware and host commands

This RP2040 image runs on BIGTREETECH SKR Pico V1.0. It controls yaw and pitch through integrated TMC2209 X/Y drivers, checks normally closed stop/limits, verifies VM and driver registers, and executes bounded jogs or periodic trajectories. It commands no laser, wire feeder or rotator motor.

Use wiring, commissioning and camera learning.

Install and flash

The released image is pgfun-positioner-r1.uf2. Its build receipt binds source and geometry hashes, compiler/SDK versions and binary/UF2 digests. With motor supply disconnected and motors unplugged, use the manufacturer's BOOT/USB procedure to mount RPI-RP2, copy that UF2, then remove the boot jumper and reconnect normally. Do not flash another controller merely because it exposes a boot disk. No device was flashed during release preparation.

Use the existing CAD Python environment, or create a separate Python 3.11+ virtual environment and install `host/requirements.txt`. The Mac capture helper lives in `tools/gun-positioner-observation/helper/`; rebuild it with `build.sh --adhoc` only if the app bundle is absent or source changed. Allow camera access to that app for physical capture.

```
tools/cad-venv/bin/python firmware/src_pgfun_positioner/host/positioner.py --list
tools/cad-venv/bin/python firmware/src_pgfun_positioner/host/positioner.py \
  --port /dev/cu.YOUR_POSITIONER --log /tmp/pgfun-console.jsonl
```

The console accepts `status`, `drivers`, `recover`, `clear`, `reference central`, `arm`, `jog yaw 32 500`, `jog pitch -32 500`, `disarm`, `stop`, `quit`. Physically align the parking pins with power off and remove them before `reference central`. Software does not measure that datum. `arm` starts a 100 ms host heartbeat. Console exit, STOP, faults and reset invalidate it.

Every jog is <=256 counts/axis and 100..2000 ms. Command rate/acceleration bounds still apply, so a short duration can reject a large jog. Travel is +/-1777 counts per axis. Client target moves are segmented without changing those controller limits.

Periodic replay

LOAD2, sequential KNOT records and PLAY load a 16..256-knot periodic Catmull-Rom path. Its 8..120 s period, cubic control envelope, rate and acceleration are checked before playback. A fresh released pedal is required. The existing rotator pedal's NO signal starts the path after 20 ms debounce and 200 ms settling. Releasing it stops playback at its current counts. Host heartbeat continues while waiting and running.

The host workflow learns responses in all four approach-direction branches, fits a smooth runout profile and accepts it only after two independent dry captures. Replaying a profile marked unqualified is refused. Geometry, response model, target, speed, direction and tube index are bound to the record. There is no actual tube-angle encoder; measured dry repeatability must qualify the existing time-based rotation.

Build and checks

Use Raspberry Pi pico-sdk **2.2.0**, Arm GNU **14.2.rel1**, CMake 3.21+ and Ninja. Their absolute installation paths are supplied by the builder's environment. Build outside the source directory:

```
python3 firmware/src_pgfun_positioner/make_geometry.py
cmake -S firmware/src_pgfun_positioner -B .cache/pgfun-build -G Ninja \
  -DPICO_SDK_PATH=/absolute/path/pico-sdk \
  -DPICO_TOOLCHAIN_PATH=/absolute/path/arm-gnu-toolchain/bin
cmake --build .cache/pgfun-build
python3 firmware/src_pgfun_positioner/make_uf2.py \
  .cache/pgfun-build/pgfun_positioner.bin \
  firmware/src_pgfun_positioner/assets/pgfun-positioner-r1.uf2
c++ -std=c++17 -Wall -Wextra -Werror -Wno-misleading-indentation \
  -I firmware/src_pgfun_positioner firmware/src_pgfun_positioner/tests/test_policy.cpp \
  -o /tmp/pgfun-policy-test
/tmp/pgfun-policy-test
tools/cad-venv/bin/python -m unittest discover -s firmware/src_pgfun_positioner/tests
```

Policy tests exercise real pulse counts, bounds, continuous replay, pedal release, independent faults, UART CRC/framing and watchdog liveness. Python tests use synthetic observations to check fitting, independent holdouts, trajectory envelopes and stale-data refusal. These results verify software behaviour; they are not physical gun-positioning records.

Print manifest

The STLs carry their bed orientation, including the optional lens shim. Solid-equivalent mass excludes supports, brim and purge.

Part / material	Size X x Y x Z (mm)	Solid g	Supports
bridge / PET-CF17	302 x 96 x 28	657.1	No
bench-toe-1 / PET-CF17	40 x 20 x 36	36.8	Yes
bench-toe-2 / PET-CF17	40 x 20 x 36	36.8	Yes
motor-pedestal / PET-CF17	100 x 94 x 62.4	181.1	Yes
yaw-bearing-cartridge / PET-CF17	86 x 114 x 80.7	169.3	Yes
yaw-output-adapter / PET-CF17	47.989 x 47.995 x 8	12.4	No
yaw-journal / PET-CF17	75.983 x 75.991 x 37.2	63.1	Yes
yaw-bearing-cap / PET-CF17	75.983 x 75.991 x 2	7.1	No
fork / PET-CF17	94 x 154 x 236	609.2	Yes
pitch-output-adapter / PET-CF17	47.989 x 47.995 x 8	12.4	No
pitch-journal / PET-CF17	75.983 x 75.991 x 37.2	63.1	Yes
pitch-bearing-cap / PET-CF17	75.983 x 75.991 x 2	7.1	No
yaw-outer-spacer / PET-CF17	51.888 x 51.894 x 7	2.6	No
yaw-inner-spacer / PET-CF17	43.99 x 43.995 x 7	2.4	No
pitch-outer-spacer / PET-CF17	51.888 x 51.894 x 7	2.6	No
pitch-inner-spacer / PET-CF17	43.99 x 43.995 x 7	2.4	No
gun-retainer-1 / PET-CF17	16 x 62 x 27.4	18.7	Yes
liner-1-lower / TPU	20.694 x 36.191 x 16	1.1	No
liner-1-upper / TPU	18.395 x 36.082 x 16	1.0	No
gun-retainer-2 / PET-CF17	16 x 62 x 27.4	19.1	Yes
liner-2-lower / TPU	20.252 x 35.716 x 16	1.1	No
liner-2-upper / TPU	18.133 x 35.688 x 16	1.0	No

Print manifest

Part / material	Size X x Y x Z (mm)	Solid g	Supports
lower-cradle / PET-CF17	147 x 62 x 30	82.1	Yes
left-cheek / PET-CF17	246.97 x 150.983 x 27.7	271.9	Yes
right-cheek / PET-CF17	246.97 x 89.983 x 30	227.2	Yes
bottom-frame-tie / PET-CF17	64 x 40 x 18	57.4	No
opposite-bearing-cap / PET-CF17	43.99 x 43.995 x 4	5.5	No
axle-collet-spacer / PET-CF17	17.996 x 17.998 x 8	1.5	No
opposite-inner-spacer / PET-CF17	17.996 x 17.998 x 8	1.5	No
axle-front-stop-spacer / PET-CF17	17.996 x 17.998 x 21	3.9	No
axle-middle-spacer / PET-CF17	17.996 x 17.998 x 12	2.2	No
axle-end-spacer / PET-CF17	17.996 x 17.998 x 18	3.3	No
axle-keeper / PET-CF17	19.996 x 19.998 x 3	1.1	No
yaw-limit-holder--1 / PET-CF17	26 x 26 x 43	5.8	Yes
yaw-limit-holder-1 / PET-CF17	26 x 26 x 43	5.8	Yes
pitch-limit-holder--1 / PET-CF17	26 x 26 x 43	5.8	Yes
pitch-switch-standoff--1 / PET-CF17	26 x 14 x 30	14.3	No
pitch-limit-holder-1 / PET-CF17	26 x 26 x 43	5.8	Yes
pitch-switch-standoff-1 / PET-CF17	26 x 14 x 30	14.3	No
umbilical-post-base / PET-CF17	90 x 80 x 190	392.1	Yes
umbilical-post-top / PET-CF17	90 x 80 x 200	407.0	Yes
umbilical-saddle / PET-CF17	60 x 34 x 43	38.5	Yes
yaw-stop-sector / PET-CF17	46 x 20 x 12	11.7	No
pitch-stop-sector / PET-CF17	46 x 20 x 12	11.7	No
controller-case / PETG	124 x 90 x 34	90.7	Yes
controller-lid / PETG	124 x 90 x 3	36.9	No
fan-guard / PETG	40 x 38 x 2	3.0	No
camera-riser-base / PETG	270 x 170 x 160	334.3	Yes

Print manifest

Part / material	Size X x Y x Z (mm)	Solid g	Supports
camera-riser-top / PETG	270 x 255 x 52.102	645.1	Yes
camera-lens-upright / PETG	125.367 x 68.673 x 108	85.1	Yes
camera-lens-cassette / PETG	108 x 86 x 17.7	154.7	No
camera-lens-retainer / PETG	76 x 76 x 2.4	11.7	No
camera-lens-front-pad / TPU	53.188 x 53.194 x 0.2	0.1	No
camera-lens-shim-0.4 / TPU	53.185 x 52.121 x 0.4	0.1	No